



Understanding the EFHW: Design • Configuration • Performance

PRESENTED BY
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About Me:

- Licensed at age 13 (Tech Plus)
- Returned to amateur radio in 2024
- Mostly CW and portable operations (POTA)
- Enjoy experimenting with portable antennas
- YouTube
- CRNA – Obstetrical anesthesia

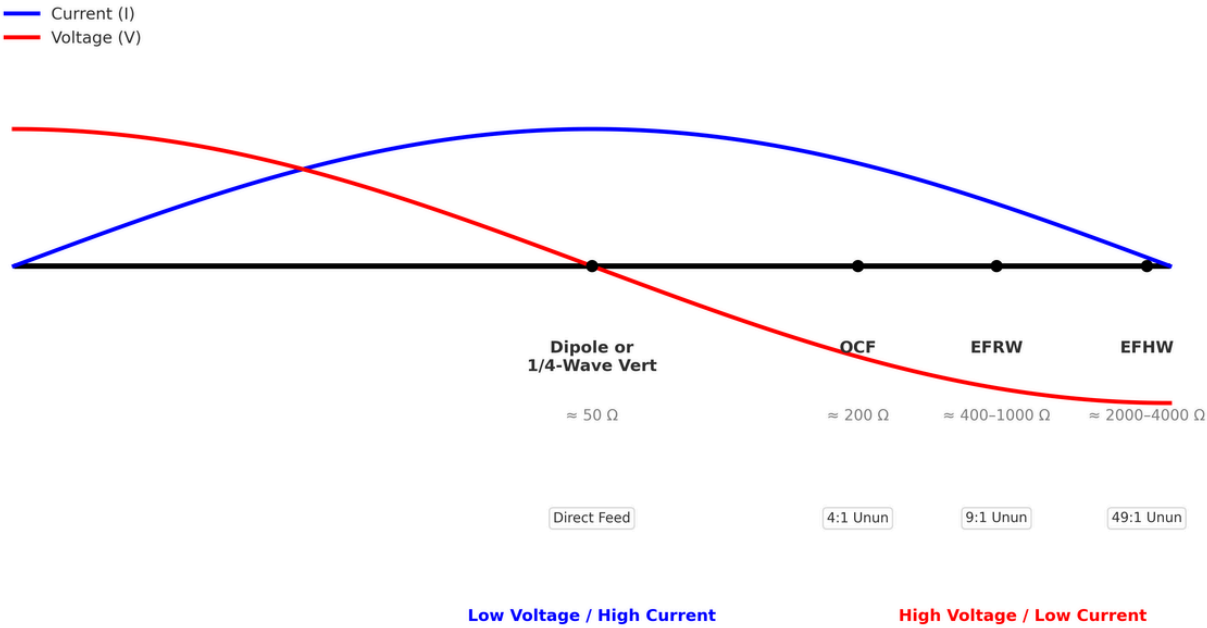
What We'll Cover

- EFHW Basics
- Feedpoint Impedance
- Transformers (49:1, 64:1)
- Counterpoise
- Deployment Options & Radiation Patterns
- Performance & Tradeoffs

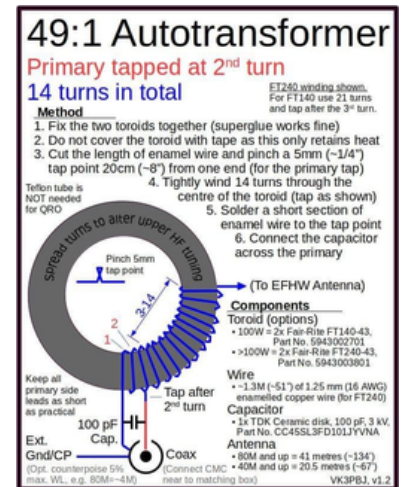
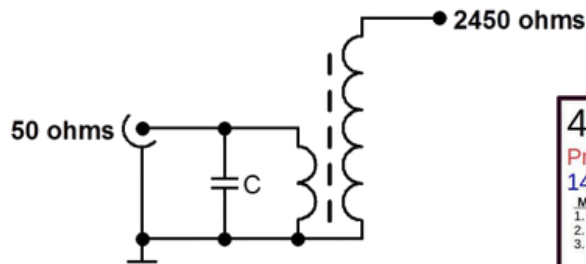
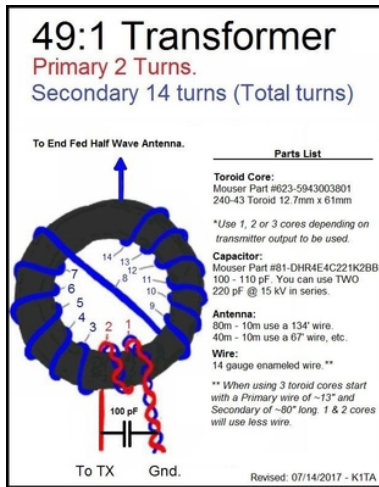
EFHW

- Popular
 - Simple – one radiating wire
 - Portable – easy to deploy
 - HOA
- Half-wave wire fed at the end
- High feedpoint impedance ($\sim 2\text{--}4\text{k}\Omega$)
- Uses impedance transformer (49:1, 56:1)
- Resonant on multiple bands

Feedpoint Impedance & Matching Along a Half-Wave



Impedance Transformers



Turn Ratio vs Impedance

Common Ununs

Turn Ratio

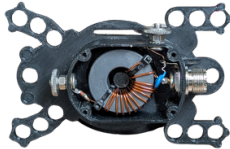
14:2 = 7:1

Impedance

$(7:1)^2=49:1$

Unun	Turns	Impedance	Windings	Use
4:1	2:1	4:1	16/8	OCF / Vert
9:1	3:1	9:1	27/9	EFRW
49:1	7:1	49:1	14/2	EFHW
56:1	~7.5:1	56:1	15/2	EFHW

Transformers Continued



Band	Frequency	Measured S21 (dB)	Loss per winding (dB) = S21/2	Efficiency (%)
80m	3.5 MHz	-0.856	-0.428	90.61%
40m	7 MHz	-0.825	-0.4125	90.94%
20m	14 MHz	-0.812	-0.406	91.08%
17m	18 MHz	-0.813	-0.4065	91.06%
15m	21 MHz	-0.853	-0.4265	90.65%
12m	24MHz	-0.99	-0.495	89.23%
10m	28 MHz	-1.131	-0.5655	87.79%



\$45

Insertion Loss:

-0.96 dB

Efficiency:

80.1%



\$89

Insertion Loss:

-0.40 dB

Efficiency:

91.2%

Fair-Rite 2643251002: 100W

Fair-Rite 2643625002: ~20w:

Wire lengths

Total Length (Feet):

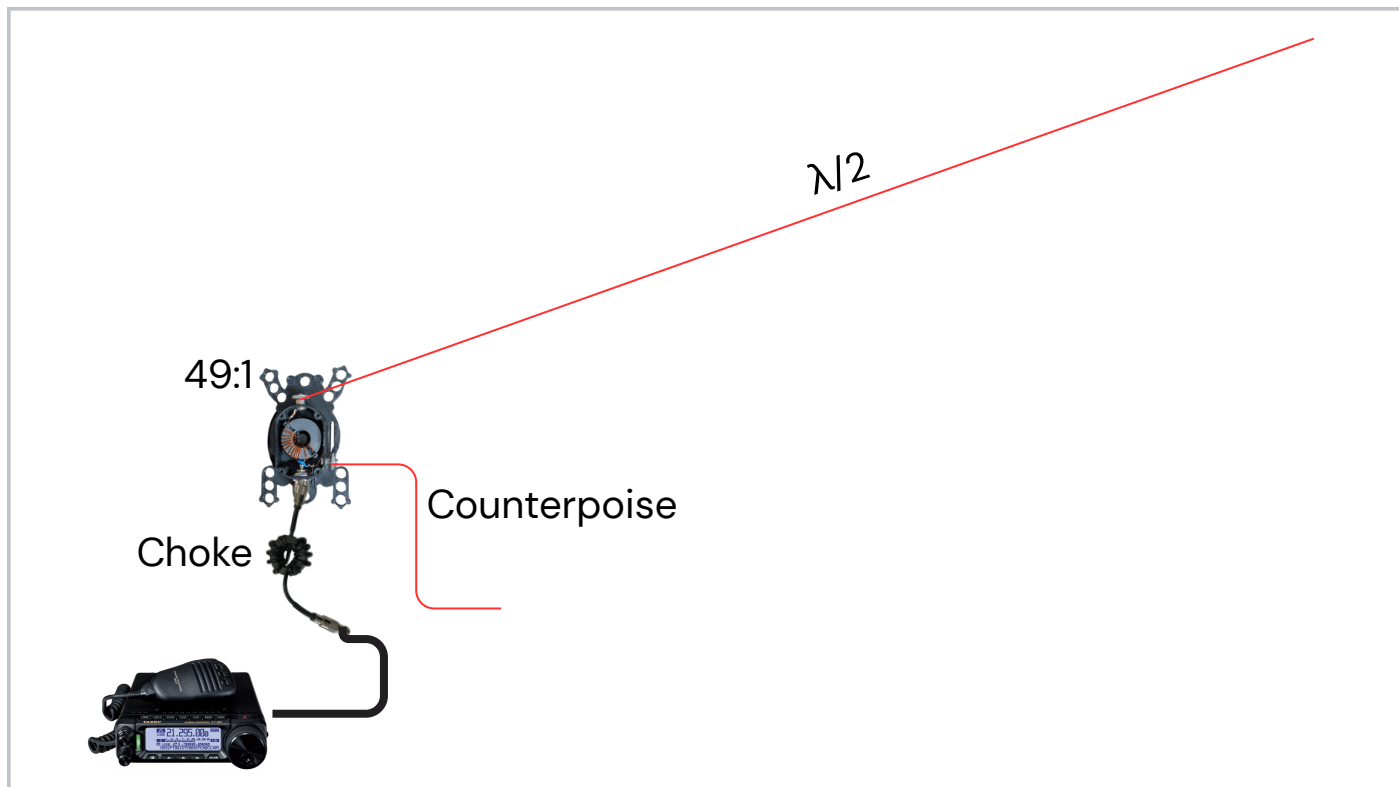
$$\frac{468}{f (Mhz)}$$

Lowest Band	Approx Length	Typical Bands
80 m	~130 ft	80, 40, 30, 20, 17, 15, 12,
40 m	~66 ft	40, 20, 15, 10
30 m	~46 ft	30, 15, 10
20 m	~33 ft	20, 10
17 m	~26 ft	17, 12
15 m	~22 ft	15

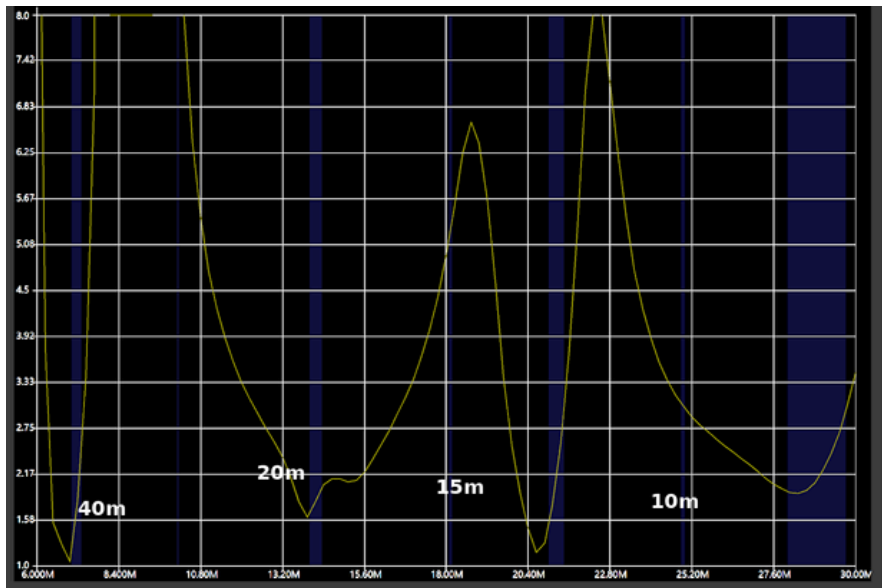
Cut wire long and trim or fold

Counterpoise?

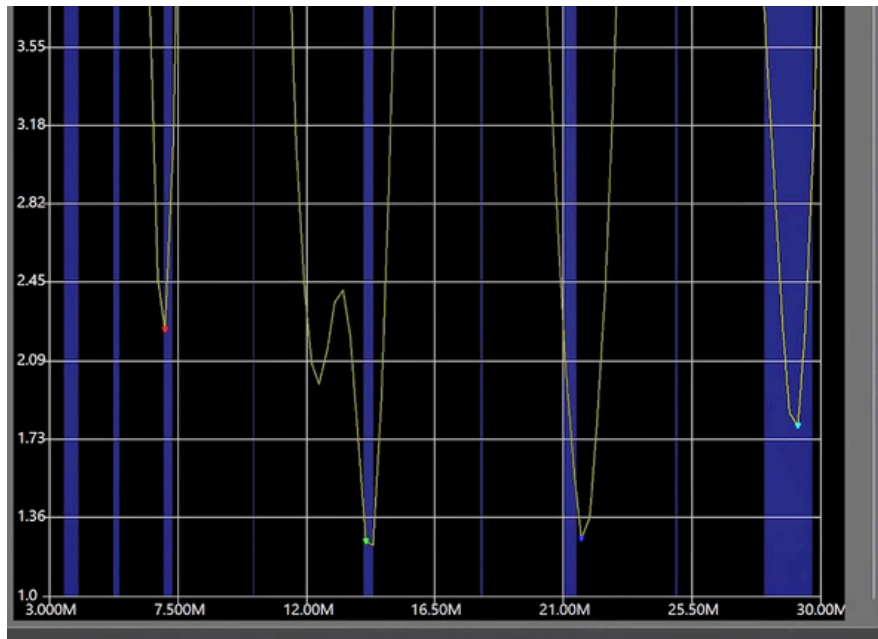
- Needed?
- EFHW still needs a return path
- Counterpoise can be:
 - short wire ($\sim 0.05\lambda$)
 - or the coax shield
- Choke helps:
 - reduce common mode current
 - stabilize SWR
 - reduce noise



SWR Sweeps



SWR Sweeps



Compensation Coil

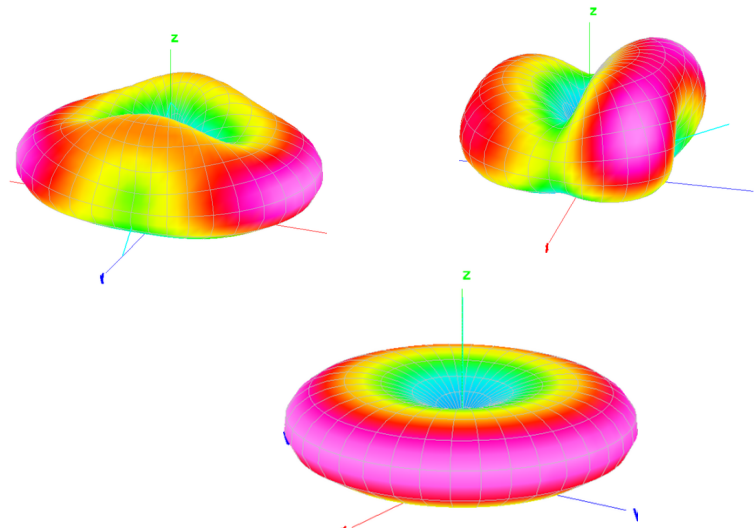


- Helps fine tune the harmonics
- Placed 6ft from feedpoint
- 5-8 turns on 1-1.25in form
- 200 pf Capacitor + 33k resistor (bleed static)
 - 30-40% up the wire

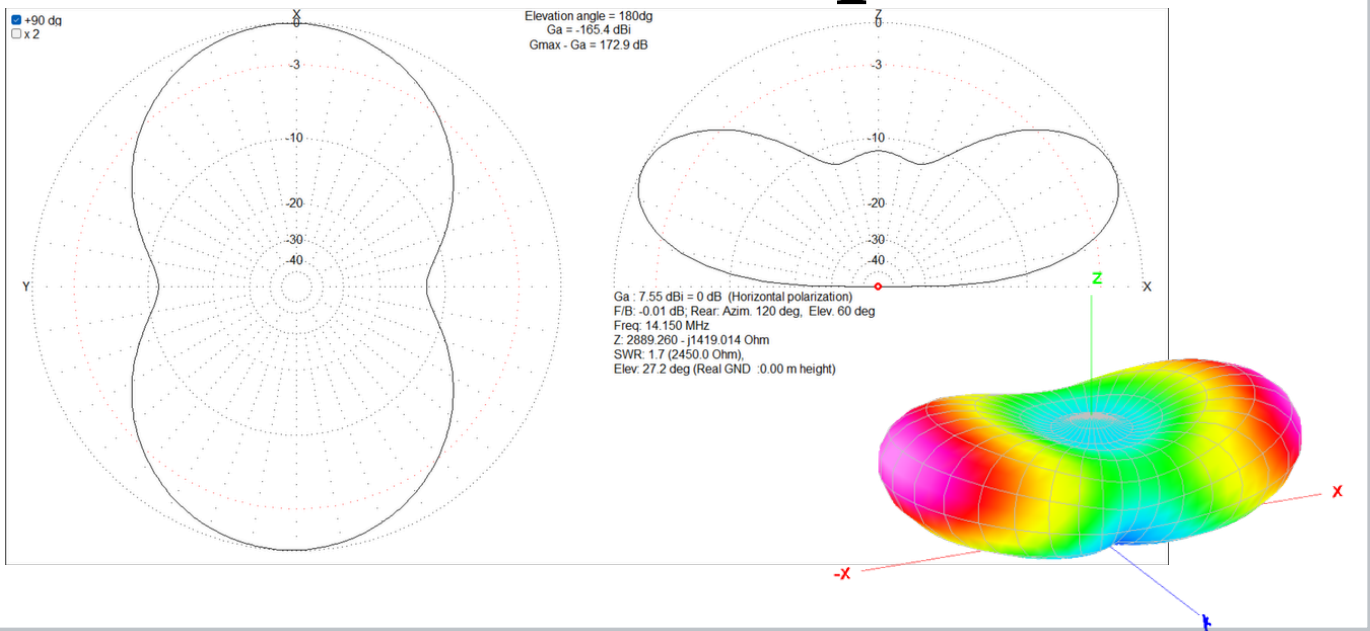


Radiation Patterns:

- Horizontal
- Vertical
- Sloper
- Inverted V
- Inverted L

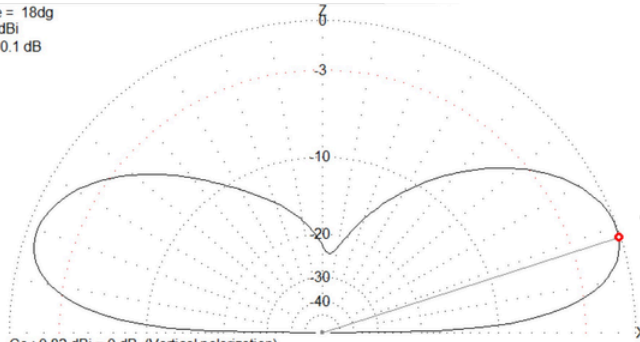


End Fed Dipole

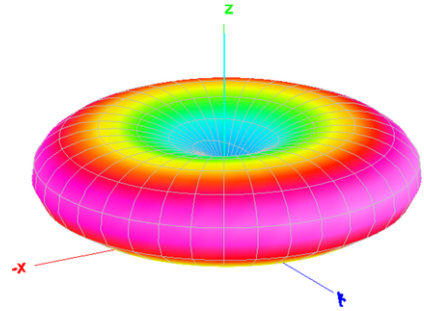


Vertical

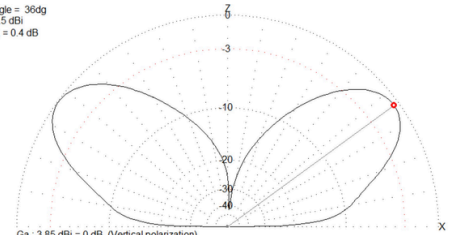
Elevation angle = 18dg
Ga = 0.8 dBi
Gmax - Ga = 0.1 dB



Ga : 0.82 dBi = 0 dB (Vertical polarization)
F/B: 0.28 dB, Rear: Azim. 120 deg, Elev. 60 deg
Freq: 14.150 MHz
Z: 2611.652 - j414.213 Ohm
SWR: 1.2 (2450.0 Ohm)
Elev. 19.9 deg (Real GND :0.00 m height)



Elevation angle = 36dg
Ga = 3.5 dBi
Gmax - Ga = 0.4 dB



Ga : 3.85 dBi = 0 dB (Vertical polarization)
F/B: -0.39 dB, Rear: Azim. 120 deg, Elev. 60 deg
Freq: 28.200 MHz
Z: 1788.390 + j669.884 Ohm
SWR: 1.6 (2450.0 Ohm)
Elev. 37.0 deg (Real GND :0.00 m height)

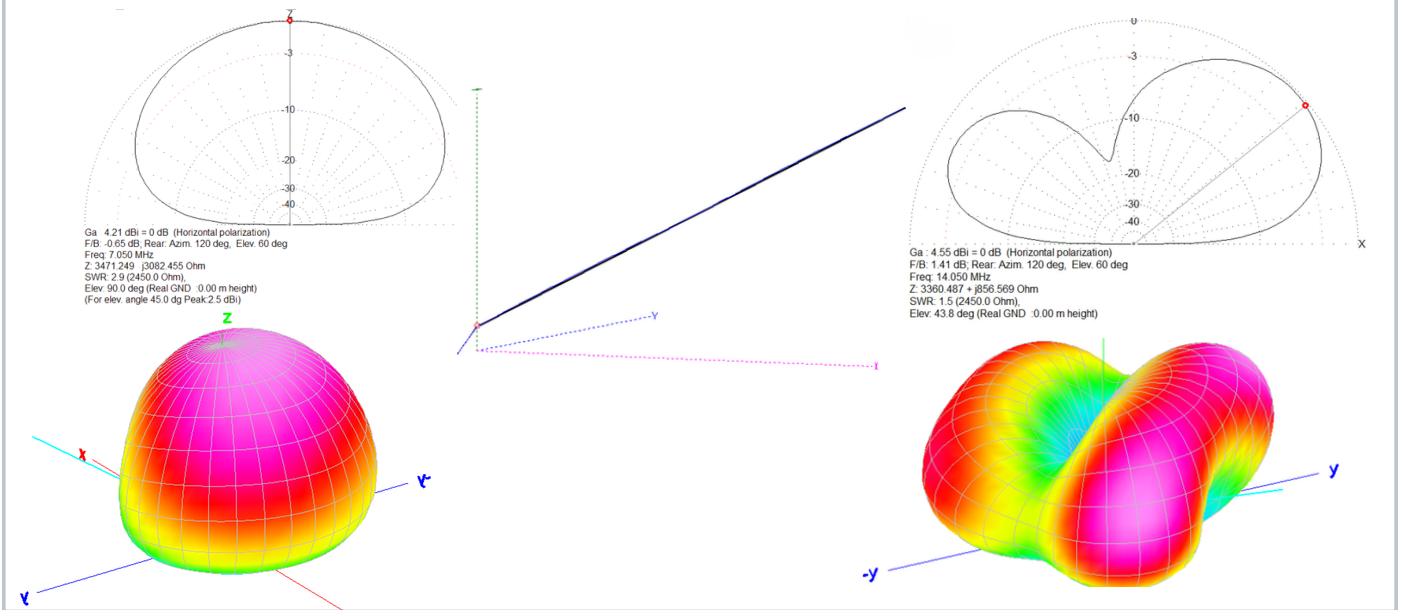
20m EFHW Vert:
0.8dBi @ 18deg
5deg - -4.6dBi

1/4 wave Vert
0dBi @ 24deg
5deg - -6.3

40M

Sloper

20M



40m mostly high angle - NVIS

function of height above ground. The patterns are going to be similar on all the configs

20m

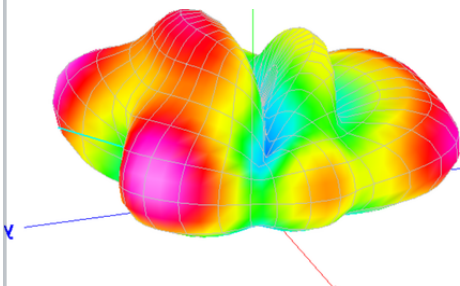
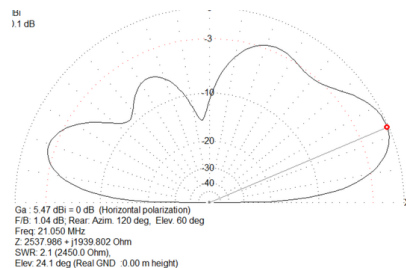
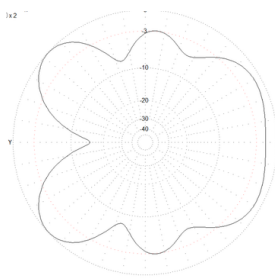
Max gain 40deg - 4.5dBi

5deg - -6.3dBi

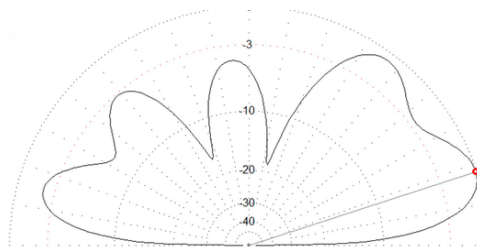
not the best DX but might work really well for POTA - state side

Sloper

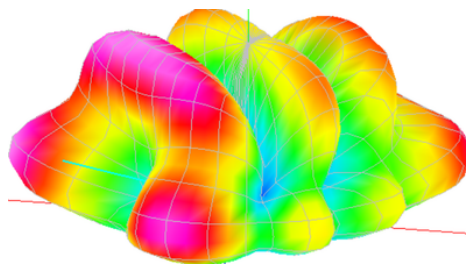
15M



10M



Ga : 5.78 dBi = 0 dB (Horizontal polarization)
 F/B: 1.42 dB, Rear: Azim: 120 deg, Elev: 60 deg
 Freq: 28.200 MHz
 Z: 3021.575 + j2504.898 Ohm
 SWR: 2.5 (2450.0 Ohm)
 Elev: 18.2 deg (Real GND: 0.00 m height)



15m:

Two main lobes 45deg off end of wire

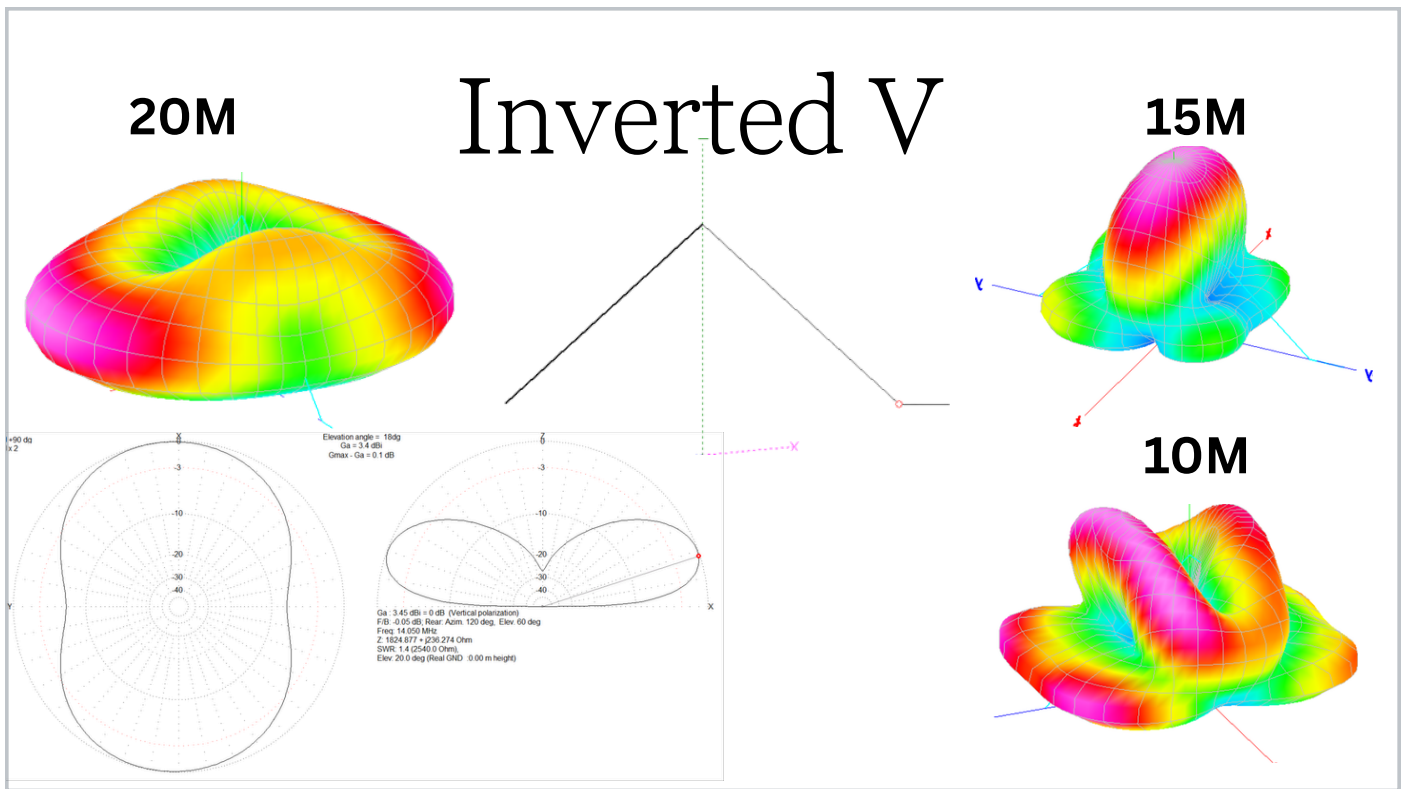
Max gain: 5.4dBi @ 24deg

at 5deg -2.3dBi

10m:

Max gain 5.8dBi @ 18deg

At 5deg 0dBi (1/4 wave -6 or -5dBi)



20m
 Broadside gain
 Low takeoff angle
 Max: 3.4dBi @ 18deg
 5deg -2dBi

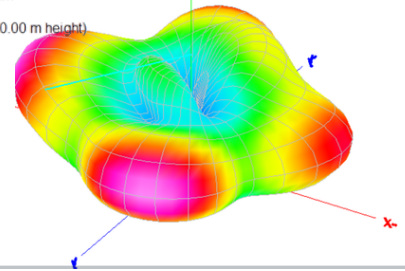
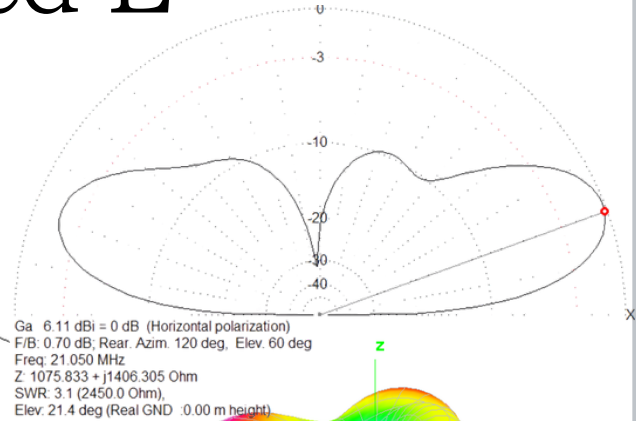
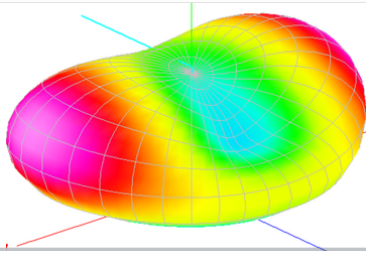
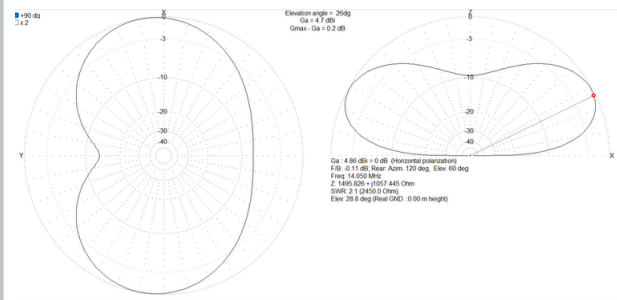
15m: not the most useful mostly high angle radiation

10m:
 A lot of high angle gain with 2 lobes lower lower angle about 40deg off the end of the wire
 Max at 11deg abt 4.2dBi

20M

Inverted L

15M



20m

Nice broad side gain with low takeoff horizontal portion

Max gain: 4.7dBi @ 26deg

5deg -3.8dBi

15m

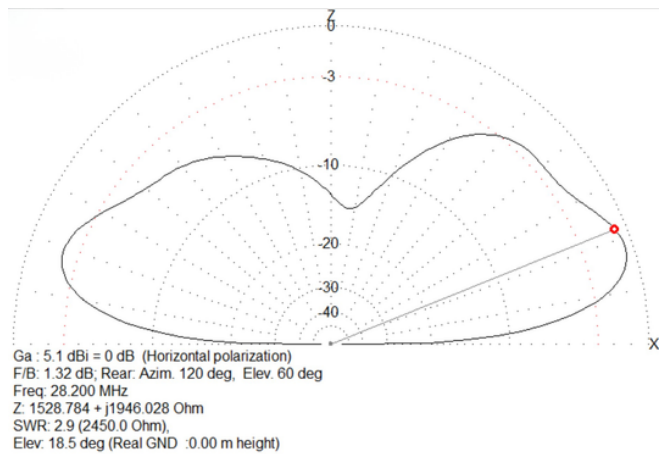
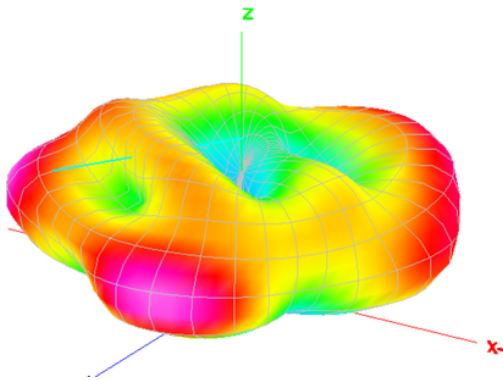
Cloverleaf with two main lobes 45deg off the end of the wire

max gain 6.1dBi @ 21deg

5deg -1.6dBi

10M

Inverted L



10m

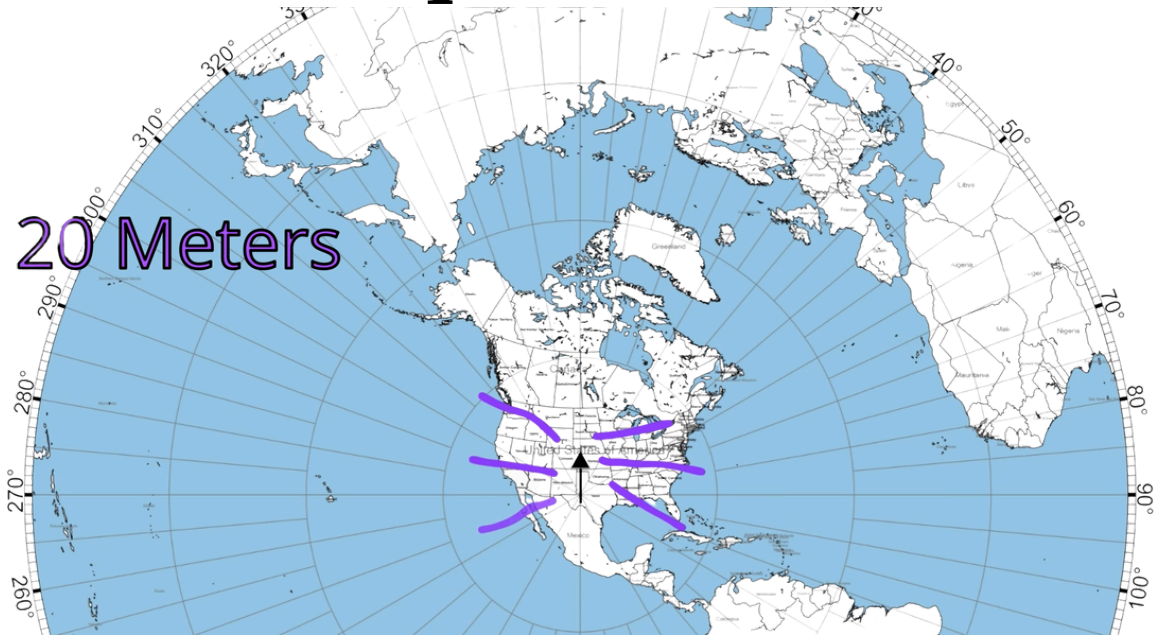
Similar radiation pattern as 15m

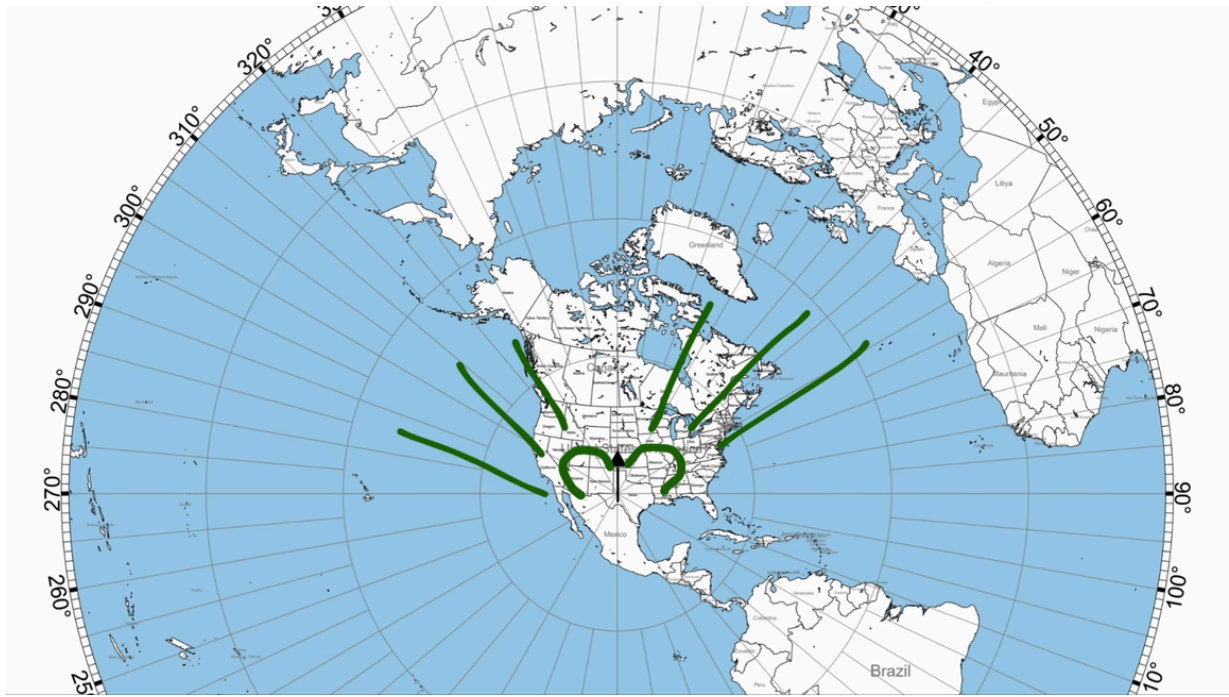
Two lobes 40-45deg off the end of the wire

Max gain is 5dBi @ 18deg

5deg -1.6dBi

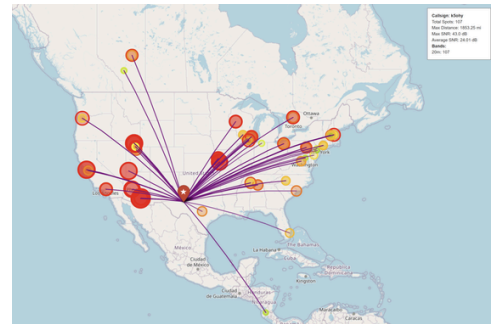
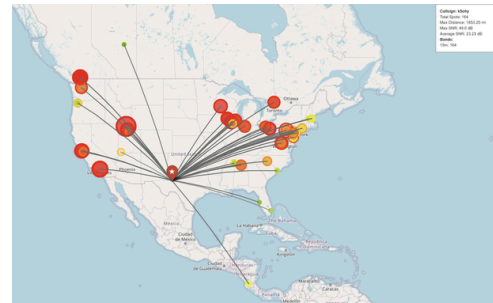
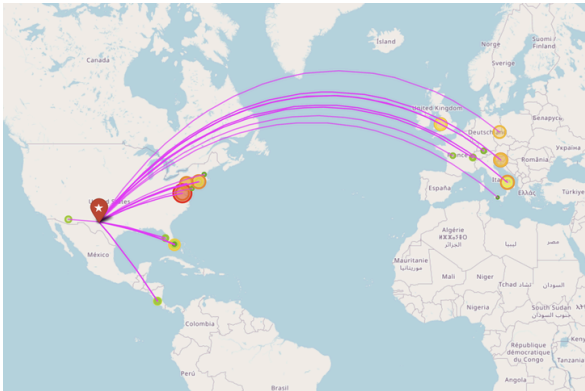
Where to point the Wire?

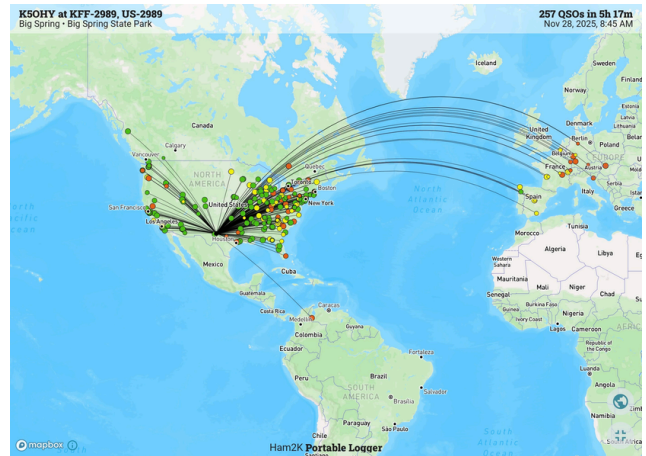
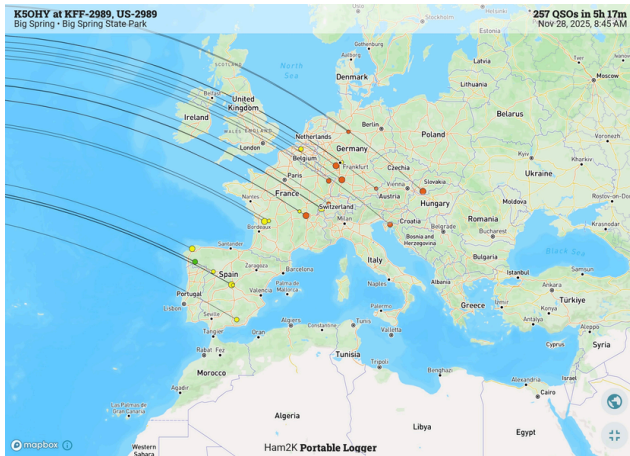






RBN Data

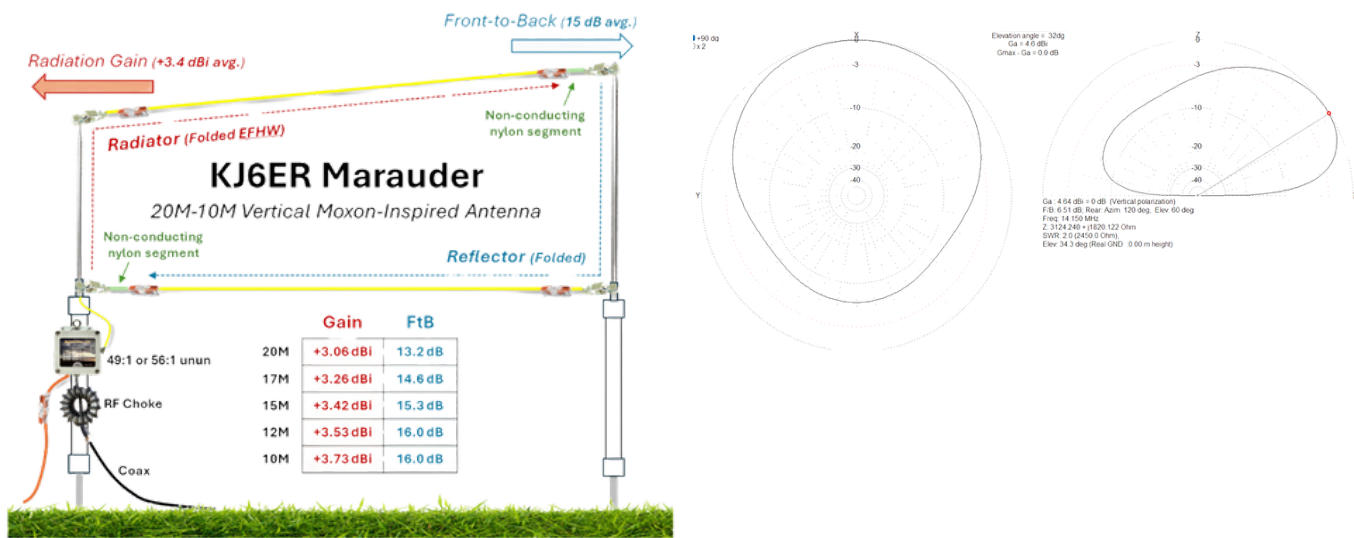




10 DXCC Entities

	United States (225)		Canada (9)		Spain (6)
	Germany (5)		France (5)		Switzerland (3)
	Slovakia		Venezuela		Croatia
	Belgium				

Other uses for EFHW



Questions?